Molecular and Cellular Biochemistry (2026) 481:1293–1308
https://doi.org/10.1007/s11010-025-05470-1

Advances in the application of spatial transcriptomics in understanding development and disease

Yao Li[1] · Qian-Wen Zheng[1] · Mingpeng Li[2] · Jing Chen[1] · Lin-Yong Zhao[3,4]

Received: 29 July 2025 / Accepted: 18 December 2025 / Published online: 9 January 2026

Abstract
Spatial transcriptomics (In this review, 'spatial transcriptomics' (ST) is employed as a overarching term, encompassing two distinct concepts. Firstly, 'tissue-level ST' refers to the capture of tissue samples at a resolution of 10–100 µm, encompassing 1–20 cells. Secondly, 'spatially resolved single-cell transcriptomics' (sc-ST) involves the analysis of individual cells or nuclei, with each sequencing unit measuring ≤ 2 µm) fuses high-throughput sequencing with positional information to map gene expression within intact tissues. By preserving spatial context, the technology uncovers cell types, signaling circuits and regulatory networks that drive organogenesis, differentiation and disease. Here we synthesize recent methodological advances and their application to developmental and clinical questions. The term "Spatial Transcriptomics" as used in this paper comprehensively encompasses all sequencing technologies that preserve spatial coordinates, including multimodal data such as transcriptomics (RNA), genomics (DNA), epigenomics (ATAC, CUT&Tag), and translationalomics (Ribo-seq).

Keywords Spatial transcriptomics · Developmental biology · Disease

Background

Tissue function arises from precise interactions between gene expression and cellular spatial organization, where cell positioning governs systemic physiological and pathological processes [1, 2]. Spatial transcriptomics (ST)—a transformative technology mapping genome-wide expression in intact tissue architectures—has become critical for decoding this spatial regulation. By combining high-throughput sequencing with spatial barcoding or imaging, ST overcomes limitations of single-cell RNA sequencing (scRNA-seq), which dissociates cells from their native contexts and erases positional information.

Modern ST platforms fall into three categories: imaging-based (e.g., smFISH, Xenium), sequencing-based (e.g., Stereo-seq, Visium), and multi-omics-integrated approaches (e.g., Spatial-CUT&Tag, RIBOmap). For example, Xenium (10x Genomics) achieves 150 nm resolution via cyclic probe hybridization, enabling subcellular RNA localization in archival FFPE samples. Stereo-seq's DNA nanoball arrays deliver whole-transcriptome profiles at 500 nm resolution, revolutionizing developmental studies [3, 4].

In developmental biology, ST has resolved long-standing questions inaccessible to traditional methods. During heart morphogenesis, ST identified NRG1 expression gradients in endocardial cells that orchestrate trabeculation via ErbB signaling—a spatial regulatory mechanism invisible to scRNA-seq (5). In oncology, ST redefined tumor microenvironment (TME) understanding by detecting PD-L1 +

Yao Li, Qian-Wen Zheng and Mingpeng Li contributed equally to this work.

✉ Jing Chen
jingchen@scu.edu.cn

✉ Lin-Yong Zhao
153795352@scu.edu.cn

1 Department of Pediatric Surgery and Laboratory of Pediatric Surgery, West China School of Medicine, West China Hospital, Sichuan University, Chengdu, China

2 State Key Laboratory of Biotherapy, West China Hospital, Sichuan University, Chengdu, China

3 Department of General Surgery & Laboratory of Gastric Cancer, West China Hospital, State Key Laboratory of Biotherapy/Collaborative Innovation Center of Biotherapy and Cancer Center, Sichuan University, Chengdu, China

4 Gastric Cancer Center, West China Hospital, Sichuan University, Chengdu, China

macrophage clusters at tumor-stroma interfaces, predicting immunotherapy response with 89% accuracy (AUC = 0.89 vs. 0.67 for bulk sequencing) [5]. Clinically, ST-guided TME subtyping in triple-negative breast cancer increased 2-year survival by 41% in recent trials [6], highlighting its translational potential.

Despite these advances, challenges remain:

Technical limitations: Most platforms have resolution > 1 µm, hindering synaptic-scale analysis (emerging techniques like SABER-FISH target sub-100 nm precision) [7]. Resolution requirements depend on biological unit size—~10 µm (mammalian cell diameter) suffices for cell-type mapping, while sub-micron precision is needed for subcellular questions.

Data integration: Incompatible barcoding systems complicate cross-platform integration (e.g., MERFISH vs. Visium), demanding novel computational pipelines [8].

Clinical adoption: RNA degradation in FFPE samples limits utility, driving innovations in probe-based amplification [9].

This review synthesizes ST's evolution from a niche tool to a cornerstone of spatial functional genomics, emphasizing its role in redefining developmental paradigms, advancing precision oncology, and addressing technical/analytical barriers. We also outline a roadmap for next-generation ST, focusing on multi-omics integration, clinical biomarker validation, and AI-driven spatial data interpretation.

Development and technological type of spatial transcriptomics techniques

The study of biological systems demands a spatiotemporal perspective, as cellular function emerges from the dynamic interplay of molecular activity and physical positioning. Spatial transcriptomics (ST) has revolutionized this paradigm by integrating histological architecture with genome-wide expression profiling, overcoming the limitations of bulk RNA sequencing (lacking cellular resolution) and single-cell RNA-seq (losing spatial context). Early spatial methods like in situ hybridization (ISH) and immunohistochemistry (IHC) were constrained to analyzing < 10 targets per run, limiting system-wide insights [4, 10].

The foundational ST protocol, introduced by Salmén et al. in 2016, preserved spatial RNA localization using barcoded microarray slides [5]. In this approach, tissue sections were placed on slides patterned with 100 µm microwells containing unique spatial barcodes. Oligonucleotides with poly-T tails captured mRNA, enabling cDNA synthesis and sequencing while retaining positional data. Though groundbreaking, this method's resolution (100 µm) restricted single-cell analyses, prompting rapid technological evolution.

Modern ST platforms now achieve subcellular precision through three complementary strategies:

Imaging-based ST (e.g., Xenium, CosMx SMI) employs cyclic probe hybridization and machine learning decoding, attaining 150–300 nm resolution in FFPE tissues. Xenium's 2023 clinical validation demonstrated 95% concordance with IHC for PD-L1 scoring in lung cancer biopsies [6].

Sequencing-based ST (e.g., Stereo-seq, Visium) utilizes spatial barcoding on DNA nanoball arrays or bead chips. Stereo-seq's 500 nm resolution recently mapped glioblastoma invasion routes by tracing *EGFR*-amplified tumor cell migration [7].

Multi-omics ST integrates epigenetics (Spatial-CUT&Tag) and proteomics (Spatial-CITE-seq). For example, Spatial-CITE-seq identified CD8 + T cell exhaustion gradients within 50 µm of PD-L1 + macrophages in melanoma, predicting anti-PD1 response (AUC = 0.87) [8].

Clinically, ST has transcended research to guide therapeutic decisions. A 2024 Phase II trial used ST-based tumor microenvironment subtyping to stratify triple-negative breast cancer patients, doubling immunotherapy response rates (45% vs. 22% with standard care) [9]. In neuropathology, ST revealed Alzheimer's-associated *APOE4* astrocyte polarization loss, correlating with tau spread velocity ($r = -0.71$) in postmortem cohorts [11].

Despite these advances, challenges persist in resolution-cell size mismatches (e.g., neuronal synapses require < 100 nm precision), data integration across platforms, and clinical standardization. Emerging solutions include CRISPR-based in situ sequencing [12] and AI-driven spatial data harmonization tools like PASTE [13]. (Figure 1)

Modern ST achieves single-cell resolution via integration with scRNA-seq, enabling precise mapping of cellular heterogeneity in intact tissues:

Imaging-based platforms (MERFISH, seqFISH) use combinatorial barcoding and error-correction to profile thousands of RNA species at 150–300 nm resolution. MERFISH recently resolved astrocyte subtype zonation in the human hippocampus, identifying GFAP+populations near amyloid plaques in Alzheimer's disease (2023). Commercial systems (Xenium, CosMx SMI) automate workflows, detecting 1,000+RNA targets per cycle in FFPE tissues.

Sequencing-based approaches (Slide-seq, Visium) leverage spatially barcoded bead arrays. Slide-seqV2 (2021) improved resolution to 1–2 µm via optimized bead density and capture efficiency, enabling single-cell analyses in pancreatic islets. A 2023 multi-center study showed 89% concordance between Visium and IHC for colorectal cancer tumor margin delineation.

Multi-omics innovations (STARmap PLUS, Spatial-CITE-seq) combine RNA detection with epigenetic/proteomic profiling. STARmap PLUS maps mRNA alongside

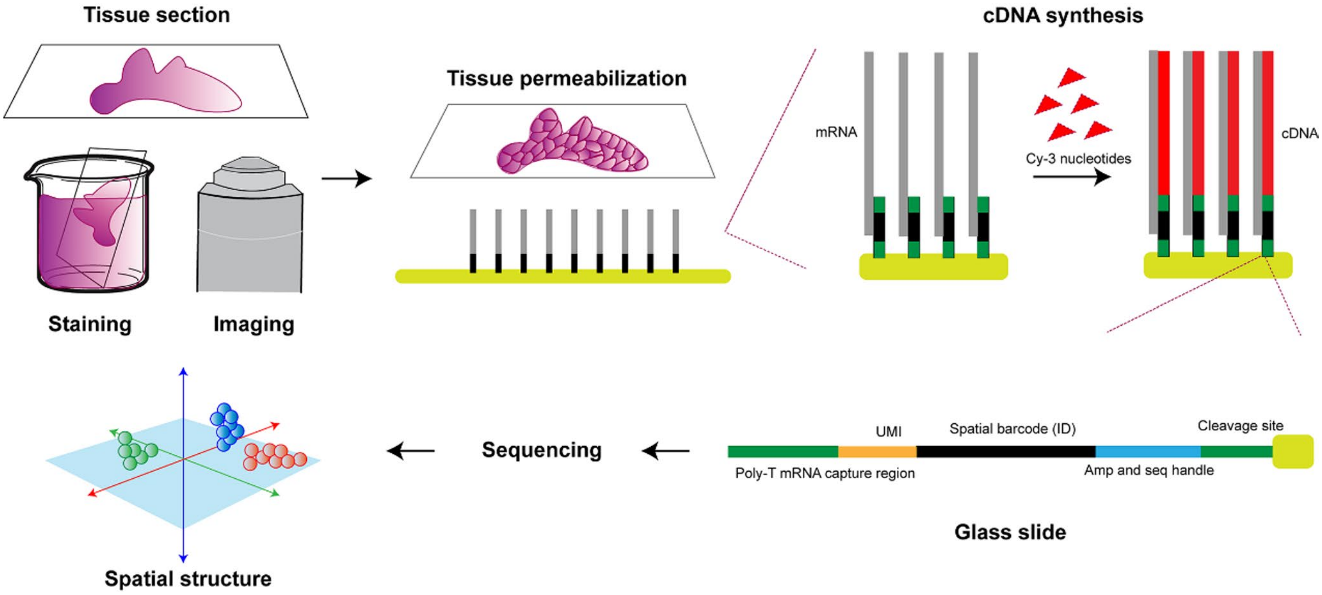

Fig. 1 General process of spatial transcriptome sequencing. In spatial transcriptomics, the tissue is sectioned, placed onto oligo(dT) primers, stained, and imaged. Next, synthesis of cDNA with Cy3-labeled nucleotides that fluoresce upon removal of tissue. Each array feature contains unique DNA-barcoded probes containing a cleavage site, a T7 amplification and sequencing handle, a spatial barcode, a unique molecular identifier (UMI), and an oligo(dT) VN-capture region, cDNA (red) is generated from captured mRNA by reverse transcription. And to further explore gene expression profiles in spatially defined domains within the olfactory bulb, principal component analysis or t-distributed stochastic neighborhood embedding (t-SNE) is used in machine learning algorithms for dimensionality reduction followed by hierarchical clustering

8 proteins at 1 µm resolution, revealing pTau co-localization with APOE4 + astrocytes in Alzheimer's models. Spatial-CITE-seq (co-profiling 200 + proteins/mRNAs) identified PD-L1 + macrophage niches predictive of melanoma immunotherapy response (AUC = 0.87).

These advancements (timeline in Fig. 2) highlight ST's rapid evolution from proof-of-concept to clinical utility. Cross-platform benchmarking now guides method selection, balancing resolution (imaging vs. sequencing) and multiplexing capacity (targeted vs. whole transcriptome) (Liao et al., 2024). Computational tools like Tangram and PASTE further integrate ST with scRNA-seq, reconstructing spatial gene regulatory networks at scale.

Key ST technologies

Par-seqFISH

A high-precision imaging technique resolving transcriptomes at 50–100 nm (single-molecule resolution) via combinatorial barcoding. Primary probes targeting 16 S rRNA with unique flanking barcodes enable multiplexed detection of > 10,000 transcripts per cell (false-positive rate: 0.04%). Its ribo-tagging system profiles microbial and host RNA simultaneously, valuable for studying host-microbiome interactions in diseases like inflammatory bowel disease.

Stereo-seq

Developed by Chen et al. (2021), it combines DNA nanoball (DNB) arrays with random barcode amplification for whole-transcriptome profiling at 500 nm resolution. DNBs with spatial barcodes form structured grids, capturing mRNA from thick tissue sections (up to 10 µm). Reverse transcription generates barcoded cDNA with unique molecular identifiers (UMIs) to reduce amplification biases.

Slide-seq

Introduced in 2019, it enables transcriptome-wide profiling in fresh frozen tissues at 10–20 µm resolution via barcoded bead monolayers (10 µm diameter) with unique oligonucleotides (SOLiD sequencing). Slide-seqV2 (2021) improved resolution to 1–2 µm (500% higher bead density, 2%→>15% mRNA capture efficiency), enabling single-cell analyses in pancreatic islets and mapping INS expression gradients in type 2 diabetes models [11]. Clinically, it aids tumor evolution studies.

SmFISH

Single-molecule fluorescence in situ hybridization visualizes gene transcripts and genomic positions in individual cells with high precision. Probes encode optical barcodes via

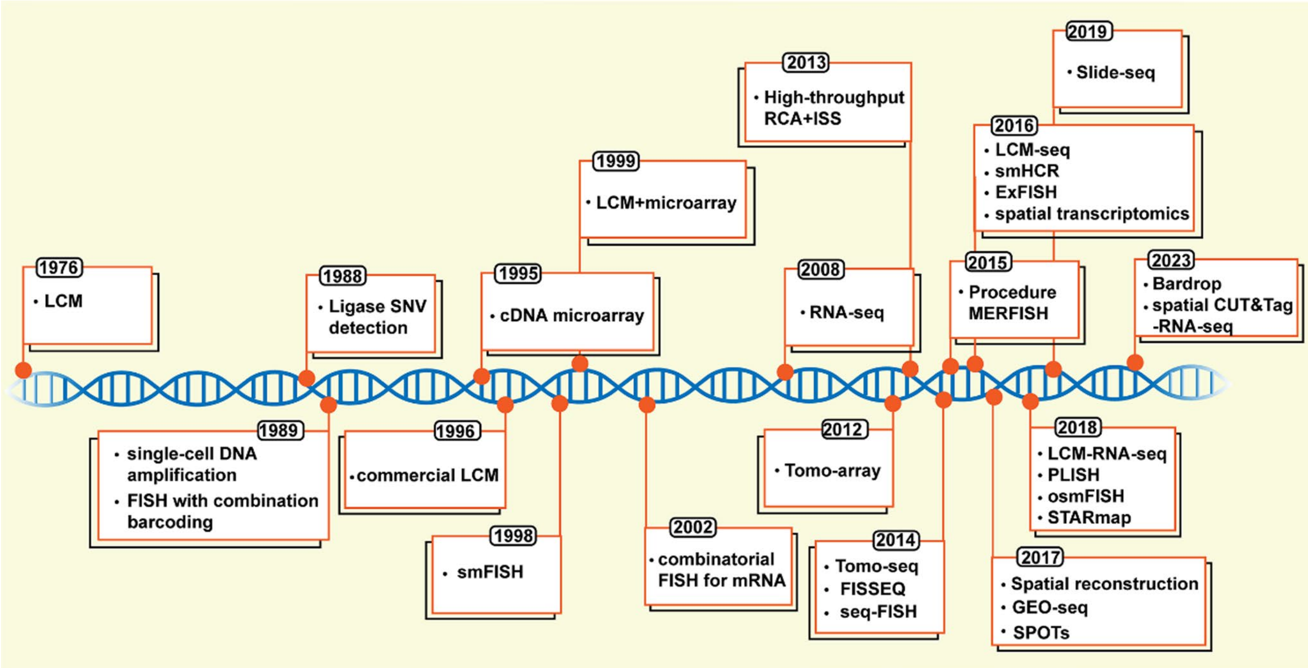

Fig. 2 The development of spatial transcriptome sequencing technology. A timeline of the birth of spatial transcriptomics and some related technologies in recent years, with the years being the birth of the corresponding technologies in the connected boxes

target-matching regions and variable detection sequences. Post-hybridization, barcodes are identified through fluorescent detection probes, high-definition imaging, and signal amplification [12, 13]. It enables single-molecule RNA detection and chromosome architecture analysis in native tissue contexts.

Spatial-CUT&Tag

Combines in situ CUT&Tag chemistry, microfluidic barcoding, and sequencing to map histone modification sites. Tissue sections are incubated with histone modification-specific antibodies, followed by secondary antibodies for pA-Tn5 transposase attachment. Transposase activation integrates adapter-laden linkers at modification sites. Microchannel-delivered DNA barcodes (Ai, 1–50) bind adapters, and complementary barcodes (Bj, 1–50) create a barcode-indexed tissue pixel grid. DNA extraction (crosslink reversal) completes library preparation.

Applied in mouse embryos, it aligns with ENCODE data for tissue-specific epigenetic regulation, elucidating cerebral cortex development and cell-type spatial distribution via histone marks. Targeting 20 µm pixels with single-cell nuclei (immunofluorescence) enables single-cell epigenomic data collection, compatible with transcriptomic/proteomic analyses for multi-omics studies.

Slide-DNA-seq

Spatially mapped bead arrays with unique DNA identifiers (sequencing ligation detection) are used. Tissue samples on beads undergo HCl treatment (histone removal) and Tn5-mediated genomic fragment generation with adapters. Spatial identifiers are released, linked to genomic fragments, and PCR-amplified for library preparation. It reconstructs tissue DNA spatial arrangement without microscopy, mapping genetic clones in mouse cancer models and human cancers. Combined with ST, it uncovers gene sets associated with clonal mutations and TME. It profiles copy number variations (CNVs) at ~1 Mb resolution, compatible with single-cell whole-genome sequencing (scWGS) for subclonal event analysis.

Spatial-CITE-seq

Developed by Rong Fan's team (Yale University), it uses antibody-derived DNA tags (ADTs) to convert protein detection into DNA sequencing, enabling spatially resolved proteomic/transcriptomic co-analysis. Tissue sections are stained with 200–300 ADTs, followed by microfluidic delivery of DNA row barcodes (A1-50) to create a 2D tissue pixel grid. NGS library sequencing (ADTs + mRNAs) reconstructs spatial protein/gene expression maps. Applicable to cancer, immunology, infectious diseases, and anatomical pathology.

DBit-seq

Uses microfluidic channels to deliver DNA barcodes to tissue surfaces. Cross-flow and in situ ligation of two barcode sets (A1-50, B1-50) create a 2D tissue pixel mosaic with unique AB barcodes (10×10 µm). It maps gene expression and protein information to precise tissue locations, aiding studies of tissue/cell spatial heterogeneity, organ formation, and multi-omics spatial sequencing.

Seq-scope

A spatially resolved barcode technology matching optical microscopy resolution, specializing in solid-phase amplification of single-molecule oligonucleotides (Illumina platform). Two sequencing phases: 1st-Seq creates a spatial barcode RNA capture grid and map; 2nd-Seq captures tissue mRNA and sequences cDNA + spatial barcodes. It maps spatial transcriptomic variations at histological and subcellular (nucleus/cytoplasm) levels, known for speed, accuracy, and ease of use.

Omni-seq/omni-ATAC

An advanced ATAC-seq iteration for chromatin accessibility profiling, improving signal-to-noise ratio and data richness. Applicable to frozen tissues and 50-micron sections. Two sequencing phases: 1st-Seq establishes a spatial barcode RNA capture grid; 2nd-Seq retrieves mRNA and sequences cDNA + spatial barcodes. It enhances chromatin accessibility map comparability, enabling applications in challenging cell lines, rare primary cells, and clinical frozen samples (Ryan Corces et al., Nature Methods).

Opto-seq

Combines optogenetic stimulation of specific inputs with single-nucleus RNA sequencing (snRNA-seq). It activates neural inputs (e.g., ventral tegmental area/VTA) via optogenetics and measures effects on neuronal subpopulations using snRNA-seq. It identifies input-activated neuronal subpopulations and analyzes immediate-early gene (IEG) expression changes, aiding complex neural network studies.

iStar

Inferring Super-resolution Tissue Architecture integrates spatial transcriptomics with high-resolution histological images via hierarchical image feature extraction. A pre-trained hierarchical vision transformer (HViT) extracts histological features (self-supervised learning/SSL), and a feedforward neural network predicts super-pixel gene expression (weakly supervised learning). It enhances ST gene expression resolution to near-single-cell levels and predicts gene expression from histological images alone.

Applications include: super-resolution gene expression pattern prediction, unsupervised tissue architecture annotation, dominant cell type visualization (scRNA-seq integration), cross-sample consistency validation, high-resolution eye development gene expression mapping, healthy/diseased tissue analysis (e.g., breast cancer), and automatic tertiary lymphoid structure (TLS) recognition.

RIBOmap

A highly multiplexed technique for spatial mapping of protein synthesis at single-cell/subcellular resolution. Its tri-probes strategy (Splint DNA, Padlock, primer probes) generates DNA amplification signals only when probes are proximal. Gene-specific barcodes in amplicons are decoded via in situ sequencing to reduce dynamic annealing/ligation errors.

It enables cell cycle-dependent translation analysis, subcellular mRNA translation profiling, cell cycle/organelle multimodal imaging, and specific gene expression capture. Applied in mouse brain tissue, it reveals cell type/brain region-specific translational regulation and post-transcriptional gene regulation (spatial translatomics vs. transcriptomics comparison).

DSP (digital spatial profiling)

A multiplexed detection method using oligonucleotide tags linked to antibodies/RNA probes via photo-cleavable (PC) linkers. UV illumination of regions of interest (ROI, single cell→~5000 cells) releases PC oligonucleotides. It achieves single-cell resolution, detecting ~ 600 mRNA transcripts. With nCounter, it analyzes 44 proteins and 96 genes (928 RNA probes); with NGS, 1412 genes (4998 RNA probes). Applicable to biobank samples, clinical decision-making, disease diagnosis/treatment, and drug target prediction.

STARmap plus

Integrates high-definition spatial transcriptomics with protein detection in the same tissue slice. SNAIL probes identify mRNA in preserved brain sections, with enzymatic amplification generating amine-modified cDNA amplicons. Proteins are labeled with primary antibodies, and a hydrogel-tissue hybrid (AA-NHS + acrylamide copolymer) immobilizes biomolecules. In situ sequencing and SEDAL technology decode cDNA amplicon gene-specific sequences, and fluorescent staining (secondary antibodies, X-34 dye) visualizes proteins.

Applications include Alzheimer's disease (AD) model studies (mapping DAM, DAA-like cells, OPCs, oligodendrocytes around Aβ plaques) and analyzing hyperphosphorylated tau distribution in excitatory neurons. It combines spatial single-cell transcriptomic states, tissue morphology, and disease markers to study physiological/pathological mechanisms.

The spatial transcriptomics (ST) technologies outlined earlier—encompassing imaging-based (e.g., Xenium, smFISH), sequencing-based (e.g., Stereo-seq, Slide-seqV2), and multi-omics-integrated platforms (e.g., Spatial-CUT&Tag, Spatial-CITE-seq)—have addressed long-standing gaps of traditional omics tools, driving breakthroughs in both basic biology and clinical translation.

In basic biology, ST preserves native tissue architecture to unravel spatiotemporal regulation. High-resolution platforms (e.g., Stereo-seq, 500 nm) identified invisible gene expression gradients, such as NRG1 in endocardial cells that orchestrate cardiac trabeculation [10]; Slide-seqV2 (1–2 μm) mapped insulin gradients in pancreatic islets to clarify β-cell dysfunction in type 2 diabetes [11]. Multi-omics tools like Spatial-CUT&Tag further linked histone modifications to gene expression in mouse embryos, revealing chromatin-level regulation of cerebral cortex development. These advances shift research from "cell heterogeneity" to "tissue spatial functional networks," enabling 3D reconstruction of developmental dynamics (e.g., gastruloid germ-layer formation [14]).

In clinical applications, ST bridges molecular profiling and translation. In oncology, Spatial-CITE-seq detected PD-L1 + macrophage clusters at tumor-stroma interfaces (AUC = 0.87 for immunotherapy prediction [8]), and a 2024 Phase II trial used ST-guided TME subtyping to double triple-negative breast cancer immunotherapy response rates (45% vs. 22% [9]). For Alzheimer's disease (AD), ST uncovered region-specific alterations (e.g., *Bok* downregulation in vulnerable CA1 neurons [15]) and plaque-induced pathological networks. Clinically compatible platforms (e.g., Xenium, 95% concordance with IHC for PD-L1 [6]) integrate with routine pathology, aiding diagnosis and therapy stratification.

Table 2 summarizes the workflow of the aforementioned technologies.

Application of spatial transcriptomics in developmental research

Embryogenesis (initial stage of higher organisms) involves cell growth, proliferation, differentiation, pattern formation, and morphogenesis, regulated by intracellular genetic material, intercellular communication, and environmental factors [16]. Thousands of genes guide cell differentiation, with spatial development regulation being a key embryogenesis mechanism [17, 18].

Spatial development patterns of embryos

Embryo axis establishment (anterior-posterior, left-right, dorsal-ventral) forms the developmental framework [19]. Maternal gene products create concentration gradients, providing positional information to guide cell specification and organ formation via gene expression and signaling.

Early development involves continuous cell division and homogeneous blastomere formation, followed by asymmetric body axis establishment. Post-axis formation, associated cells are specified at axis-related positions, forming developmental compartments and eventually organs [19]. Positional information originates from oocytes—maternal proteins and mRNA act as morphogenetic determinants, directing differential gene expression [20].

Morphogens (maternal gene products) regulate cell differentiation and fate via gene activation/inhibition. Distributed as concentration gradients in oocytes, they trigger different gene expression at threshold concentrations. For example:

Drosophila embryos: Bicoid (anterior transcription factor) forms an anterior-posterior concentration gradient; Caudal (posterior transcription factor) forms a posterior-anterior gradient. High Bicoid/low Caudal induces head-forming genes; low Bicoid/Caudal induces chest-forming genes; high Caudal/no Bicoid induces abdominal-forming genes [20, 21].

Vertebrate embryos: Notochord acts as a morphogen source [22].

Later development involves cell-cell or cell-extracellular matrix interactions (receptor-ligand signaling) that shape cell formation and specification. Spatial cell arrangement determines signal reception, driving development [23]. (Figure 3)

Gene expression exhibits spatiotemporal dynamics:

1. Strict temporal-spatial sequence for gene activation/shutdown.
2. Selective gene expression underpins cell fate.
3. Cells in different functional states express distinct genes; abnormal expression causes diseases.
4. Chromatin 3D conformation affects gene accessibility (e.g., promoter-enhancer spatial distance regulates gene activation) [24,25].

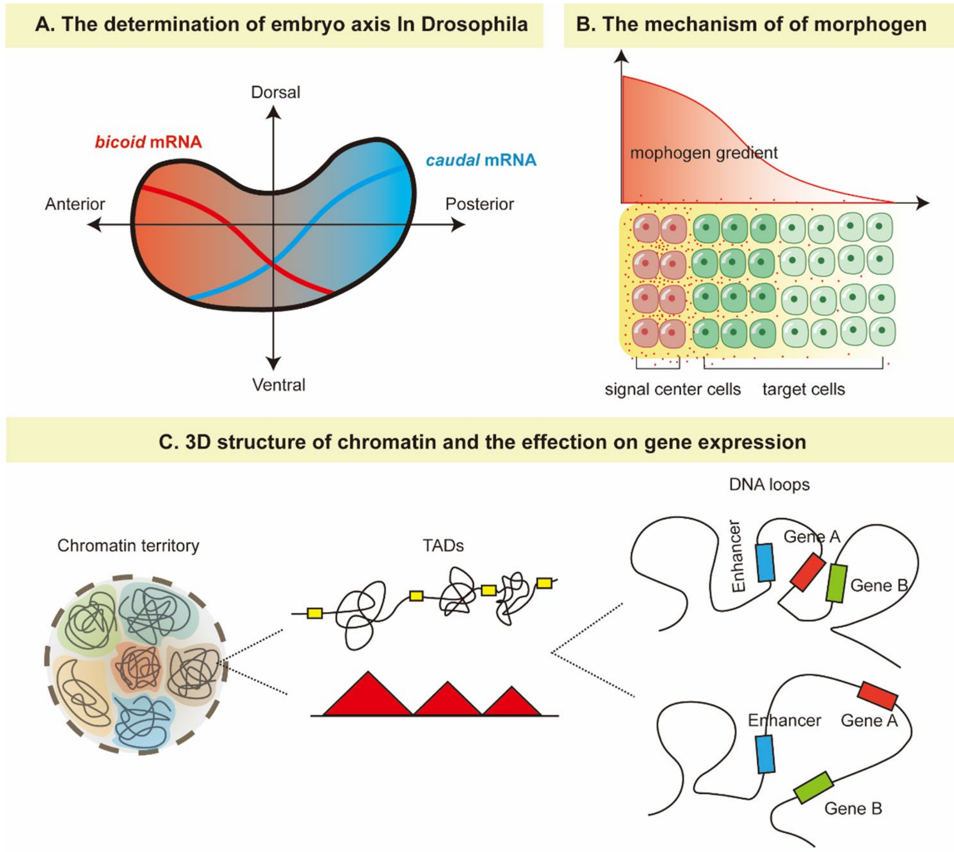

Fig. 3 Spatial development patterns of embryos. **A** The pattern of embryonic development in Drosophila. *bicoid* and *caudal* form head-to-tail and tail-to-head concentration gradients, respectively. Different concentration gradients activate different genes and guide the formation of the head, chest and tail of the embryo. **B** Regulation of morphogens. The signal center cells send morphogens with a concentration gradient in a specific direction. Cells along the axis have different developmental fates because of the different concentrations of morphogen in their locations. **C** Three-dimensional regulation of genes. Chromatin forms chromatin territory. Frequent interactions of chromatin form Topologically associating domains (TADs). The chromatin of different TADs is relatively isolated. DNA mutations may lead to the destruction of TADs, and the change of interaction pattern of chromatin resulting in abnormal gene expression

Application of spatial transcriptome in developmental biology

Single-cell RNA sequencing (scRNA-seq) captures diverse gene expression patterns across cellular stages to inform developmental processes, but its requirement for cell dissociation prevents studying tissue cells in their natural context. Integrating scRNA-seq with spatial transcriptomics (ST) enables a more holistic understanding of cellular interactions and communication. Below is recent progress in developmental biology driven by spatial transcriptome data.

Gastruloid development

Gastruloids are 3D clusters of about 300 embryonic stem cells that mimic mammalian embryo features (germ-layer differentiation, body axis formation), though their comprehensive gene expression profile was previously undefined. Comparing scRNA-seq and ST data between gastruloids and mouse embryos revealed striking similarities, confirming gastruloids accurately mimic embryonic somatic development and positioning them as a valuable tool for mammalian embryonic development research (including in vitro studies).

Heart development

The heart is the first functional embryonic organ which originates from the mesoderm, develops into four chambers by about 30 days post-conception, and differentiates into structures like the myocardium and aorta, driven by precise spatiotemporal gene expression interactions. Integrating ST, scRNA-seq, and in situ sequencing (ISS) allowed characterization of spatiotemporal gene expression variations during human heart morphogenesis at single-cell resolution, identifying shared spatial expression patterns across developmental stages and cell type heterogeneity at intermediate stages. Mouse models have also been used to investigate cell heterogeneity in embryonic and adult hearts.

Lung development

The lung's complex structure and microenvironment require ST to clarify cell assembly and disease-related cell functions. A research team analyzed healthy lungs/airways from deceased donors, mapped cell types along the respiratory tree's proximal-to-distal axis, and distinguished 80 cell types/states (including 11 unannotated in prior lung maps), generating the most comprehensive human lung cell map to date.

Spermatogenesis

Our understanding of spermatogenesis' molecular mechanisms is limited by a lack of accurate in vitro models. Chen et al. integrated Slide-seq with targeted in situ RNA sequencing to create a near-single-cell resolution spatial transcriptome map of mammalian testes, which identifies testicular cell locations, recreates seminiferous tubule architecture, and infers germ cell gene expression dynamics (leveraging spermatogenesis' cyclical, stage-specific nature). A comparison of wild-type and diabetic mouse testicular spatial maps revealed disrupted seminiferous tubule cell structure may cause diabetes-related male infertility.

Intestinal development

The intestine comprises interrelated cell types that form its unique morphology, but their developmental molecular basis is unclear. Amelia T. Soderholm et al. linked scRNA-seq with ST to establish a large single-cell spatiotemporal map of human intestinal development, creating a morphogenesis map (spanning time, location, and cell compartments) and analyzing key events (epithelial crypt formation, mesenchymal differentiation, GALT emergence) [26]. ST also enabled single-cell spatial reconstruction of laser-captured villi fragments, revealing functional specialization along villus cells.

Endometrial development

The human endometrium undergoes hormone-driven periodic shedding, regeneration, and differentiation (coordinated by the hypothalamus-pituitary-ovary axis) during reproductive life. Single-cell and spatial transcriptomic analysis characterized the cellular states and spatial locations of endometrial cells in the proliferative/secretory phases of childbearing-age women. The resulting cell map can deconstruct dominant cell types in endometrial cancer and endometriosis.

Brain development

Daniel Gyllborg's team used ST on adult mouse brains to generate a comprehensive molecular brain atlas—constructed solely by classifying tissue gene expression patterns into anatomical definitions. This approach uncovered previously unrecognized brain regions/subregions with layer-specific characteristics [27]. Additionally, enhanced in situ sequencing (ISS), built on proximity ligation assays (PLPs) and rolling circle amplification (RCA), has been used to investigate spatial features of human brain coronal sections and whole mouse brains [28].

Application of spatial transcriptomics in disease research

Spatial mechanism of disease occurrence and development

Tissue cell type distribution is critical for disease research (Fig. 4). The microenvironment strongly influences disease initiation and progression:

Immune system: Cytokine-immune cell interactions regulate immune responses and autoimmune diseases (e.g., rheumatoid arthritis, inflammatory bowel disease) [29].

Oncology: TME drives cancer initiation, progression, and metastasis. Cancer cells interact with normal cells and immune cells (lymphocytes, mononuclear macrophages), inspiring therapies like CAR-T, CAR-NK, and immunotherapies [30–32].

scRNA-seq identifies novel cell types and heterogeneity but loses cellular spatial context, altering gene expression patterns post-dissociation. ST pinpoints disease-associated

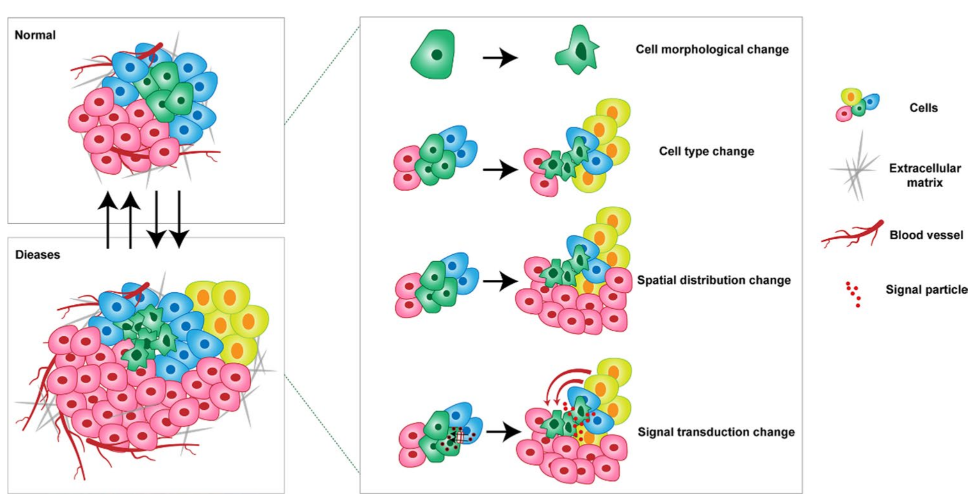

Fig. 4 Spatial mechanism of disease occurrence and development. Spatial differences in normal and pathological status include changes in cell morphology, cell type, cell spatial distribution and extracellular matrix, and changes in intercellular signaling

cell types, studies molecular mechanisms/pathways (signal transduction, metabolism, cytokines), and reveals disease progression (e.g., tumor cell subtypes for targeted therapy, inflammatory cell distribution in autoimmune diseases). Immune checkpoint inhibitors (anti-PD-1/PD-L1) are effective for high immune cell infiltration tumors [33]. In infectious diseases, pathogen-host cell interactions, pathogen load, and immune cell responses guide treatment selection.

Disease spatial development highlights interconnectedness between disease, surrounding cells, and environment (gene-environment crosstalk). ST provides a comprehensive perspective for disease research.

Application of spatial transcriptomics in disease research

ST studies disease tissue gene expression heterogeneity, disease-related cell type spatial distribution, progression mechanisms, and therapeutic potential, advancing disease understanding and treatment.

Tumors

Tumor evolution involves dynamic molecular/phenotypic changes in cancer cells (dedifferentiation, drug resistance, poor outcomes) driven by genetic/epigenetic alterations, transcriptional variations, and treatment pressure—contributing to inter/intratumor heterogeneity [34, 35]. ST applications include:

Prostate cancer: Distinguishes healthy or diseased areas more accurately than pathological annotations, revealing gene expression changes during progression [36].

Pancreatic ductal carcinoma: ST and scRNA-seq identifies spatially enriched ductal cells, macrophages, dendritic cells, and cancer cells, clarifying inflammatory fibroblast-cancer cell co-localization (stress response gene expression) [37].

Lung adenocarcinoma (LUAD): ST and multiplex IHC analyzes subtype molecular features and cell plasticity, describing transcriptional reprogramming and hypoxia-induced regulatory networks. Macrophage subtyping aids tumor immunity research [38].

Skin squamous cell carcinoma: scRNA-seq and ST reveals four tumor subsets (three resembling normal epidermal states, one immunosuppressive cancer-specific keratinocyte population) [39].

Breast cancer: SCSubtype (single-cell method) identifies recurrent tumor cell heterogeneity; ST uncovers stromal-immune cell interactions, aiding anti-tumor immune regulation research [40].

Nervous system diseases

Alzheimer's disease (AD): ST analysis of AD mouse brains identifies stress response/transcription regulatory gene groups in the hippocampus and olfactory bulb, and spatial Bok downregulation (mitochondrial physiology/cell death) in mouse/human AD hippocampi [15]. Early-stage: Myelin/oligodendrocyte gene (OLIG) co-expression networks change around amyloid plaques; late-stage: Complement system, oxidative stress, lysosome, and inflammation networks are altered [41].

Amyotrophic lateral sclerosis (ALS): ST measures gene expression in mouse spinal cord and human autopsy tissues, analyzing temporal-spatial molecular events driving motor neuron-glia interactions and motor neuron loss [42].

Septicemia

A dynamic disease requiring precise spatiotemporal definition for biomarkers and targets. ST and scRNA-seq (mouse sepsis model) analyzed kidney cellular/molecular events, identifying endothelial/stromal cells as early responders and cell-cell communication failure as the main cause of endotoxemia organ dysfunction [43].

Osteoarthritis

ST explores inflammatory characteristics in arthritis joint biopsies. Rheumatoid arthritis (RA) features adaptive immediate response profiles and T-B cell interactions; spondyloarthritis (SpA) involves tissue repair-related functions [44]. It deepens understanding of chronic inflammatory disease cellular mechanisms and tissue diversity.

Periodontitis

ST quantified and localized gene expression in periodontitis-affected gingival tissue, identifying significantly upregulated genes in inflammatory connective tissue (vs. non-infected tissue) [45].

Mental disorders

Depression's pathophysiology remains unclear. ST analysis of control, resilient, and susceptible depression mouse models annotated eight brain regions and six cell types, revealing significant transcriptional changes in the hippocampus, isocortex, and amygdala—highlighting their roles in disease vulnerability and resilience [46].

Unique contributions of spatial transcriptomics (ST) in applied research

ST preserves gene expression spatial context, revealing biological phenomena inaccessible to conventional techniques:

Decoding tissue microenvironments at cellular Resolution

Unlike scRNA-seq (tissue dissociation disrupts spatial relationships), ST maps cell-type distributions in native niches. For example:

Human intestinal tissues: AI-powered STASCAN delineated cell layer boundaries, identifying unrecognized gene expression gradients (FOXM1, MYBL2, SLC12A2) regulating nutrient absorption and barrier function.

Breast cancer: ST uncovered immune-tumor cell interactions and immune-excluded regions explaining therapeutic resistance.

Reconstructing developmental dynamics in 3D

ST + algorithms like Spateo generate 3D "molecular holograms" of developing tissues. For example:

Mouse embryos: Spateo linked macroscopic morphological changes (heart ventricular curvature) to gene expression programs, identifying asymmetric organ growth regulators.

Mouse brain: STAM (stereotaxic topographic atlas) achieved 1 μm resolution, identifying 236 new regions via 3D reconstruction.

Unraveling spatial heterogeneity in disease

AD: Stereo-seq mapped hippocampal single-cell resolution, identifying CA1 neuron vulnerability and CA4 neuron resilience. Amyloid-β plaques induced distinct gene networks (microglia: C1qa/C1qb; oligodendrocytes: OLIG downregulation), explaining global amyloid therapy failure.

Oncology: ST revealed immune checkpoint molecule spatial gradients in TME, with immune "cold spots" correlating with poor prognosis—enabling precise targeting.

Integrating multi-omics with spatial context

ST combines gene expression data with histological imaging. For example:

Human lung tissues: ST detected micron-scale pulmonary fibrosis structures, linking TWEAK-Fn14 signaling to histological abnormalities.

Cross-species analysis: Spateo revealed vertebrate heart complexity via spatial reorganization of conserved gene modules.

Technical advancements

Resolution enhancement: STASCAN fills capture domain gaps, achieving subcellular resolution.

Cost-effective 3D mapping: Spateo's virtual slices reduce physical sections, lowering costs.

Multi-modal integration: STARmap PLUS combines spatial transcriptomics with protein detection at subcellular resolution. (Table 1)

Challenges and prospects

Technical challenges

Limitations in core technology, including performance, sample compatibility, and experimental workflows:

Insufficient spatial resolution and sequencing coverage: Mainstream platforms (e.g., 10x Genomics Visium) have 100–200 μm spot resolution (multiple cells per spot), constrained by tissue section thickness, chip capture array size, and RNA capture efficiency.

Low detection sensitivity: Mixed signals from multiple cells dilute low-abundance RNA (especially in high-cell-density tissues), hindering rare cell type identification and low-abundance functional gene detection (e.g., cell fate-regulating transcription factors).

Limited transcript capture scope: Most techniques capture single-ended (not full-length) transcripts, impeding immune cell receptor profiling and alternative splicing analysis.

Poor FFPE sample compatibility: FFPE samples are clinically abundant, but ST applicability remains limited (compatible methods under development).

Complex experimental workflows: Specialized steps (precise tissue slicing, spatial labeling, chip-based capture) increase complexity and error risk (e.g., tissue fixation quality, section thickness control).

Analytical challenges

Difficulties in data processing, interpretation, integration, and tool/expertise limitations:

Large and complex data:

High volume: Integrating gene expression and spatial context requires deep sequencing, increasing storage/processing demands.

Mixed/noisy signals: Multi-cell spot signals complicate cell-type-specific gene expression disentanglement; high noise and multidimensional heterogeneity hinder normalization, cross-sample integration, and pattern mining.

Table 1 Representative studies of Spatial transcriptomics on development of organs and disease

Species-Tissue	Disease	Tissue Regions	Key Cell Populations & Markers	Refs
Human/Mouse-Gastruloid	Healthy	Germ layers (endoderm, mesoderm, ectoderm) and body axes (anterior-posterior, dorsal-ventral)	Embryonic stem cell-derived germ layer cells; Markers: Mesoderm (T/Brachyury), Endoderm (Sox17), Ectoderm (Sox2)	[14]
Human/Mouse-Heart	Healthy	Ventricles, atria, heart tube, outflow tract (critical for myocardium/septum/aorta development)	Cardiomyocytes (MYH6, MYH7), Endothelial cells (CD31), Mesenchymal cells (VIM), Epicardial cells (WT1)	[47–50]
Human-Lung	Healthy	Proximal-to-distal axis of respiratory tree (trachea, bronchi, bronchioles, alveoli)	Alveolar epithelial cells (AT1: AGER; AT2: SFTPC), Bronchial epithelial cells (SCGB1A1), Goblet cells (MUC5AC), Ciliated cells (FOXJ1), 11 previously unannotated cell populations	[51]
Mouse-testis	Healthy; Diabetes	Seminiferous tubules (stage-specific regions), interstitial regions	Spermatogonia (PLZF), Primary spermatocytes (SYCP3), Secondary spermatocytes (TDRD1), Round spermatids (PRM1), Sertoli cells (SOX9), Leydig cells (CYP11A1); Diabetes-associated cells with disrupted seminiferous tubule structure	[52]
Human-Intestinal tract	Healthy	Intestinal villi, crypts, mucosal layer, muscular layer, vascular system, GALT (Gut-Associated Lymphoid Tissue)	Intestinal epithelial cells (EPCAM), Goblet cells (MUC2), Paneth cells (LYZ), Enteroendocrine cells (CHGA), Mesenchymal cells (PDGFRα), Immune cells (CD45, CD3, B220)	[26]
Human-Endometrium	Healthy; Endometriosis; Endometrial carcinoma	Proliferative/secretory endometrium (functional/basal layers); Ectopic endometrial lesions (endometriosis); Tumor foci & paracancerous tissue (endometrial carcinoma)	Endometrial epithelial cells (KRT19), Stromal cells (VIM), Immune cells (CD68, CD3, CD19), Cancer cells (Ki-67, p53)	[53]
Human/Mouse-Brain	Healthy	Mouse brain anatomical subregions (hippocampus, cortical layers, amygdala); Human brain coronal sections (prefrontal cortex, hippocampus)	Neurons (NeuN, MAP2), Astrocytes (GFAP), Microglia (IBA1), Oligodendrocytes (OLIG2), Layer-specific cells (e.g., cortical layer V: CTIP2 + neurons)	[27, 28]
Marmoset-Gastrulation	Healthy	Gastrula germ layers (ectoderm, mesoderm, endoderm), primitive streak, Hensen's node	Gastrula germ layer cells; Markers: Mesoderm (Brachyury), Ectoderm (Sox2), Endoderm (GATA6)	[54]
Mouse-Bone marrow	Healthy	Hematopoietic sinuses, hematopoietic islands, stromal regions	Hematopoietic stem cells (CD34+, Sca-1+, c-Kit+), Granulocytes (Gr-1+), Macrophages (F4/80+), B cells (B220+), T cells (CD3+), Stromal cells (PDGFRβ+)	[55]
Human/Mouse-Kidney	Healthy; septicemia	Renal cortex (glomeruli, proximal/distal tubules), renal medulla (collecting ducts, loops of Henle), renal interstitium	Healthy: Podocytes (NPHS1), Renal tubular epithelial cells (AQP1, AQP2), Endothelial cells (CD31); Septicemia: Endothelial cells (CD31), Stromal cells (VIM), Macrophages (CD68)	[56–58]
Human-Skin	Healthy; Psoriasis; Dermatitis; Keloids; Squamous cell carcinoma	Healthy: Epidermis (basal/spinous/granular layers), dermis; Lesions: Psoriasis (hyperkeratosis zone), Keloids (collagen deposition zone), Tumor nests (squamous cell carcinoma)	Keratinocytes (KRT14, KRT10/KRT16), Immune cells (CD4+/CD8 + T cells, CD11c + dendritic cells), Cancer cells (KRT5, p63), Fibroblasts (α-SMA)	[39, 59–62]
Human-Pancreas	Healthy	Pancreatic islets, exocrine pancreas (acini, ducts)	Islet β-cells (INS), α-cells (GCG), δ-cells (SST), Acinar cells (AMY2A), Ductal cells (KRT19)	[63]
Human/Mouse-Brain	Healthy; Alzheimer's disease	Hippocampus (CA1/CA3 regions), olfactory bulb, cerebral cortex (Aβ plaque-surrounding zones)	Healthy: Neurons (NeuN), Astrocytes (GFAP); AD: Disease-associated microglia (DAM: CD68+, TREM2+), Disease-associated astrocytes (DAA-like: GFAP+, AQP4-), Oligodendrocytes (OLIG2), APOE4 + astrocytes, pTau + neurons	[15, 41, 42]
Human-synovial tissue from knee or hip joints	Rheumatoid arthritis; spondyloarthritis	Synovial lining/sublining layers, inflammatory infiltration zones	RA: T/B cells (CD3+, CD19+), Macrophages (CD68+), Fibroblast-like synoviocytes (FLS: α-SMA+); SpA: Tissue repair-related cells, Inflammatory cytokine-producing cells (TNF-α+)	[44]

Table 1 (continued)

Species-Tissue	Disease	Tissue Regions	Key Cell Populations & Markers	Refs
Human-Gingival tissue	Healthy; Periodontitis	Gingival epithelium (basal/spinous layers), gingival connective tissue (inflammatory/healthy zones)	Epithelial cells (KRT14), Fibroblasts (VIM), Immune cells (CD45+, CD11b+, CD3+), Inflammatory cytokine-producing cells (IL-6+, TNF-α+)	[45]
Mouse-Brain	Depression	Hippocampus, isocortex, amygdala	Neurons (NeuN, c-Fos + activated neurons), Astrocytes (GFAP), Microglia (IBA1), BDNF + neurons (depression-related transcriptional changes)	[46]
Human-Kidney	Healthy; Diabetic kidney disease	Renal cortex (glomeruli, tubules), renal interstitium (fibrotic zones in diabetes)	Podocytes (NPHS1, WT1), Renal tubular epithelial cells (AQP1, AQP2), Interstitial fibroblasts (α-SMA, COL1A1), Macrophages (CD68+)	[64, 65]
Human/Mouse-Granulomas	Granulomas	Granuloma core, peripheral inflammatory zone, fibrotic zone	Macrophages (CD68+, CD14+), Multinucleated giant cells (CD68+), T cells (CD3+, CD4+, CD8+), Fibroblasts (α-SMA)	[66]
Human-Liver	Healthy; Primary sclerosing cholangitis	Hepatic lobules (central vein, portal tract), bile ducts (diseased ducts in PSC)	Hepatocytes (ALB), Biliary epithelial cells (KRT19), Hepatic stellate cells (α-SMA), Macrophages (CD68+), T cells (CD3+)	[67]
Human-Skin	psoriasis vulgaris; acute; chronic atopic dermatitis; lichen planus	Psoriasis (parakeratosis zone); Atopic dermatitis (epidermal spongiosis, dermal inflammation); Lichen planus (interface dermatitis, hyperkeratosis)	Keratinocytes (KRT16, KRT17), Th17 cells (IL-17+, CD4+), Mast cells (TPSAB1+), Immune cells (CD11c + dendritic cells)	[68]
Human-Liver	Liver fibrosis	Hepatic lobules (injury/fibrotic zones), portal tract	Hepatic stellate cells (α-SMA, COL1A1), Hepatocytes (ALB, CYP3A4), Macrophages (CD68+, TREM2+), Biliary epithelial cells (KRT19)	[69]
Human-White adipose tissue	Healthy	Adipose lobules (white adipocyte zones), perivascular regions, interstitial regions	White adipocytes (PPARγ, ADIPOQ), Vascular endothelial cells (CD31), Mesenchymal stem cells (PDGFRα+), Immune cells (CD45+, CD11c+)	[70]
Human/Mouse-Lung	Healthy; Idiopathic pulmonary fibrosis	Healthy: Alveoli, bronchioles; IPF: Fibrotic foci (honeycomb lung, fibroblast foci)	Healthy: AT1 cells (AGER), AT2 cells (SFTPC); IPF: Fibroblasts (α-SMA, COL1A1), Myofibroblasts (POSTN), Macrophages (CD68+)	[71, 72]
Human-Lung	Healthy; Idiopathic pulmonary arterial hypertension	Pulmonary arterioles (diseased/normal arterioles), alveolar regions	Pulmonary arterial endothelial cells (CD31, eNOS), Smooth muscle cells (α-SMA, SM22α), Macrophages (CD68+), CD4 + T cells	[73]
Mouse-Lung	Diffuse alveolar damage	Damaged/normal alveoli, pulmonary interstitial inflammatory zones	Alveolar epithelial cells (SFTPC, AGER), Endothelial cells (CD31), Neutrophils (Ly6G+), Macrophages (F4/80+), IL-6 + inflammatory cells	[74]
Human-Ligament	Healthy; Degenerated	Ligament fiber bundles (normal/degenerated fractured fibers), perivascular regions	Ligament fibroblasts (SCX, TNMD), Inflammatory cells (CD68+, CD3+), Vascular endothelial cells (CD31)	[75]

Missing low-expressed gene information: Limited sequencing depth requires external data integration (e.g., multi-omics) or supplementary algorithms.

Inadequate bioinformatics tools: Most tools are designed for RNA-seq/scRNA-seq, lacking spatial feature and data structure analysis capabilities.

High expertise thresholds: Requires interdisciplinary backgrounds (statistics, cell biology, immunology), limiting widespread adoption.

Clinical challenges

Barriers to clinical translation, primarily cost and scalability:

Excessive costs: Higher than conventional transcriptomics (sample preparation, sequencing, analysis).

Infeasible large-scale clinical cohorts: High costs and time-consuming workflows restrict large-sample studies (critical for biomarker validation and disease mechanism research), limiting translation to personalized medicine and diagnosis.

Prospects

Resolution improvement: New platforms with higher-density capture chips/probes reduce cell coverage per spatial spot (e.g., ~50 μm spatial bands). Integration of ST with scRNA-seq (isolating single cells from spatial slices) preserves spatial information while enhancing resolution (e.g., Slide-seq, Slide-seqV2). Super-resolution microscopy + ST data integration improves cellular localization accuracy.

Table 2 The workflow of the technology mentioned in the text

Technology	Image generation	Tagging principle	Coordinate lattice
par-seqFISH	Low-magnification autofluorescence mosaic (anatomical reference) + high-NA confocal imaging (fluorescent amplicons)	Primary padlock probes (20-nt gene-specific region); RCA for amplicons; fluorescent reverse oligos for decoding	single-cell level (no preset physical grid; spatial identity is determined a posteriori from the centroid of each segmented cell).
Stereo-seq	Post-tissue apposition: TIRF fluorescence sequencing imaging + bright-field histological image	DNBs with 25-nt coordinate barcode + 30-nt oligo(dT); mRNA capture → in-situ barcoded cDNA (with UMI)	tissue-grid level (barcodes are pre-deposited in a hexagonal array with 220 nm pitch).
Slide-seq	Pre-library: Haematoxylin-stained cryosection bright-field/H&E image; post-sequencing: bead coordinate-histology alignment	10-µm beads with 8-nt spatial barcode; mRNA capture → reverse transcription (append barcode, SOLiD sequencing)	tissue-grid level (beads are randomly packed but their x–y positions are pre-decoded via fluorescent sequencing).
smFISH	Wide-field/light-sheet microscopy z-stack; max-intensity projection → 2D anatomical map	~48 fluorescent 20-nt oligos (target mRNA); Gaussian fitting for spots; 3D centroid recording	single-cell level (each detected spot is mapped to the nearest segmented cell).
Spatial-CUT&Tag	Post-CUT&Tag: H&E image (same slide); affine transformation (sequencing map-H&E alignment)	Histone Ab recruits pA-Tn5; transposase activation (adapter integration); microchannel barcodes (Ai/Bj) for modification sites	tissue-grid level (barcode spots are photolithographically defined at 10 µm pitch).
Slide-DNA-seq	Pre-library: Low-magnification DAPI image (nuclei); post-sequencing: bead coordinate-DAPI alignment	Random N9 primer (replace oligo(dT)); HCl (histone removal) → Tn5 fragmentation; genomic fragments + bead barcode	tissue-grid level (randomly packed 10 µm beads whose positions are pre-decoded).
Spatial-CITE-seq	Pre/post-assay: Tissue fluorescence scan; Ab fluorescence channel (data alignment)	ADTs (protein labeling); microfluidic delivery of A1-50 row barcodes; ADT + mRNA co-sequencing (jointly barcoded library)	tissue-grid level (100 µm antibody spots define the positional code).
DBit-seq	Post-two orthogonal flow-striping: Bright-field image (10-µm stripes as fiduciary grid)	Universal oligo(dT) (mRNA capture); x/y-direction 25-nt barcodes; in-situ RT (append 50-nt composite barcode)	tissue-grid level (deterministic 10 µm grid).
Seq-Scope	Post-in-situ sequencing: High-res (×40) autofluorescence mosaic (tissue alignment)	Epoxy-coated coverslip with 0.5-µm pitch amine-modified barcoded primers; mRNA capture → in-situ RT (retain sub-micron barcode)	tissue-grid level (0.5 µm pitch pseudo-array).
Omni-seq	Pre-assay: H&E image; post-assay: Cy3 fluorescence (photobleached grid); dual-channel registration	Photo-cleavable 30-nt barcode (50-µm squares); UV release (≤5 µm diffusion); oligo(dT) hydrogel (mRNA capture)	tissue-grid level (DMD-projected 50 µm squares).
Opto-seq	Pre-laser: Live nuclear GFP fluorescence; post-capture: bright-field (tissue integrity check)	Membrane-tethered Tn5 (405-nm laser activation); ROI-specific 25-nt barcode (label DNA fragments)	single-cell level (ROI can be as small as one cell).
iStar	Pre-sequencing: Cyclic immunofluorescence (cycIF) z-stack; post-sequencing: bead coordinate-IF affine mapping	Post-IF: mild fixation + Slide-seq barcoding; mRNA capture (10-µm beads with spatial barcode)	tissue-grid level (10 µm bead array).
RIBO-map	Pre-imaging: Low-magnification DAPI (nuclear segmentation); high-NA imaging (fluorescent spot decoding)	RNase I (ribosome-protected footprints); padlock probe (20-nt gene-specific region); RCA + fluorescent probe decoding	single-cell level (no preset grid; spots are assigned to segmented cells).
Digital Spatial Profiling (DSP)	Whole-slide fluorescence scan (×20); user-drawn ROIs	Ab/ISH probe with 70-nt photo-cleavable barcode; UV release (≤50-µm aperture); nCounter/NGS (barcode quantification)	tissue-grid level (user-selected ROIs, effectively a coarse grid).
STARmap PLUS	Post-probe stripping: Two-photon z-stack; final high-res image (cell segmentation)	Padlock probe (circularization + RCA); sequential fluorescent probe decoding (amplicon sequences); 3D centroid registration	single-cell level (spatial identity is assigned post-segmentation).

Data processing advancement: AI/machine learning enhances efficiency and accuracy; user-friendly analysis platforms and integrated tools increase accessibility.

Multi-omics integration: Expanding combinations of transcriptomics, genomics, epigenomics, and proteomics for comprehensive biological insights.

Conclusions

Spatially resolved transcriptomics has emerged as a transformative technology, enabling high-resolution gene expression mapping across biological systems. Its applications span diverse samples (mouse brain, zebrafish embryos, mammalian liver, frozen breast tumors, bacterial populations) and targets (mRNA, exons, introns, lncRNAs), with resolution approaching subcellular levels and thousands of cells profiled routinely.

This technology has revolutionized tissue architecture understanding, revealing developmental and disease pathogenesis mechanisms. By identifying critical cell subtypes, decoding microenvironmental interactions, and elucidating immune dynamics in cancer/autoimmune diseases, it has become indispensable in developmental biology, clinical research, and therapeutic discovery.

Despite challenges (cost, sensitivity, sequencing depth, specialized bioinformatics), innovations in multi-omics integration, computational tools, and protocol optimization are addressing these limitations. As resolution improves and costs decline, spatial transcriptomics is poised to become a cornerstone of foundational research, drug development, and precision medicine, accelerating insights into human health and targeted therapy translation.

Acknowledgements We apologize for not being able to cite all the publications related to this topic due to space constraints of the journal.

Author contributions Lin-Yong Zhao conducted full reviews, verifying that all data, figures. Yao Li, Qian-wen Zheng and Ming-peng Li drafted the initial manuscript. Jing Chen supervised the study. All authors read and approved the final manuscript.

Funding This review is supported by the Sichuan Science and Technology Program, No.2025ZNSFSC0715. No.2021YFS0111, to Lin-Yong Zhao.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

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